

Hyperparameter Tuning and Optimization Report

Abstract

This project focuses on improving machine learning model performance through hyperparameter tuning. A Random Forest Classifier was trained on the Titanic Survival Prediction dataset and optimized using Grid Search Cross Validation. The project evaluates the impact of different hyperparameter combinations and identifies the best configuration for the model.

Introduction

Hyperparameters are configuration settings that control the behavior of machine learning algorithms. Selecting suitable hyperparameters can significantly influence model performance. This project applies Grid Search Cross Validation to optimize a Random Forest Classifier and evaluate its effectiveness on the Titanic dataset.

Dataset Description

The Titanic dataset contains passenger information and survival outcomes.

Features

- Passenger Class
- Sex
- Age
- Fare
- Number of Siblings/Spouses
- Number of Parents/Children
- Embarked Location

Target Variable

- Survived

Data Preprocessing

The following preprocessing steps were performed:

- Removed irrelevant columns such as PassengerId, Name, Ticket, and Cabin.
- Filled missing Age values using the median.
- Filled missing Embarked values using the mode.

- Converted categorical variables into numerical form using one-hot encoding.
- Split the dataset into training and testing sets.

Baseline Model

A Random Forest Classifier with default parameters was trained to establish baseline performance.

Baseline Accuracy

82.12%

Hyperparameter Tuning

Grid Search Cross Validation with 5-fold cross validation was used to evaluate different hyperparameter combinations.

Hyperparameters Evaluated

- `n_estimators` = [50, 100, 200]
- `max_depth` = [5, 10, 20]
- `min_samples_split` = [2, 5, 10]
- `min_samples_leaf` = [1, 2, 4]

Results

Best Hyperparameters

- `max_depth` = 5
- `min_samples_leaf` = 1
- `min_samples_split` = 2
- `n_estimators` = 100

Best Cross Validation Score

83.56%

Tuned Model Accuracy

81.56%

Classification Report

Class	Precision	Recall	F1-Score
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Not Survived (0)	0.81	0.90	0.85
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Survived (1)	0.84	0.69	0.76
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Overall Accuracy: 82%

Weighted F1-Score: 81%

Discussion

The optimized model achieved a cross-validation score of 83.56%, indicating strong performance across validation folds. The model demonstrated excellent recall for passengers who did not survive and maintained balanced precision and recall values overall. The results show that Grid Search is an effective method for selecting optimal hyperparameters.

Challenges Faced

- Selecting suitable hyperparameter ranges.
- Increased execution time due to multiple parameter combinations.
- Handling missing values before model training.
- Evaluating model performance across multiple validation folds.

Conclusion

This project successfully demonstrated hyperparameter tuning using Grid Search Cross Validation. The Random Forest Classifier achieved strong predictive performance, and the tuning process helped identify the most effective hyperparameter configuration. Hyperparameter optimization remains an essential step in building reliable machine learning models.

References

- Scikit-Learn Documentation
- Titanic Dataset
- Machine Learning with Python Documentation